

Steve Nwaiwu

Interdisciplinary AI Researcher | Human-Centered NLP & Multimodal Learning | Explainability, Robustness & Fairness in Socio-Technical Systems

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RESEARCH INTERESTS

Interdisciplinary research at the intersection of artificial intelligence, social systems, and public policy, including causal and explainable multimodal AI; fairness and accountability in disaster-response and high-stakes decision systems; robust and adversarial-resistant learning; misinformation and harmful content in online environments; vision–language modeling; counterfactual reasoning; and trustworthy, human-centered AI for socially vulnerable populations.

EDUCATION

Rajamangala University of Technology Thanyaburi, Thailand Jun 2024 – Dec 2025
Master of Science (Research) in Data & Information Science GPA: 4.00 / 4.00
Thesis research on robust and explainable machine learning for hate-speech detection in online discourse, incorporating counterfactual reasoning, adversarial perturbation techniques, and fairness-aware learning frameworks to address bias and model reliability in socio-political contexts.

Rajamangala University of Technology Thanyaburi, Thailand Jul 2020 – Apr 2024
Bachelor of Science (Honors) in Big Data Analytics & Management GPA: 3.33 / 4.00
Graduated with honors in a research-driven program focused on statistical learning, data pipeline architecture, and applied AI. Capstone project involved developing a centralized institutional analytics platform for real-time performance monitoring and multi-departmental decision-making, using ETL pipelines and interactive visualization.

Yaba College of Technology, Lagos, Nigeria Sep 2008 – Dec 2011
National Diploma in Computer Science GPA: 4.11 / 5.00
Built foundational proficiency in software engineering, systems design, and relational database management. Completed applied coursework and group-based projects in algorithm design, network infrastructure, and object-oriented programming—laying groundwork for later work in data-driven AI systems.

ACADEMIC EXPERIENCE

Rajamangala University of Technology Thanyaburi (RMUTT) Thailand
Lecturer / Teaching Faculty (Undergraduate Level) Jan 2024 – Present

- Developed and delivered undergraduate lectures, hands-on labs, and assessments with a focus on active learning and real-world applications in AI and programming.
- Courses taught:
 - **Computer Programming** (09151105): Designed beginner-friendly modules introducing structured and functional programming in Python, integrating Jupyter-based labs and algorithm design practices.
 - **Artificial Intelligence** (09152205): Taught foundational AI concepts, including uninformed and informed search, supervised learning, and neural networks, with a focus on ethical implications and model evaluation.
- Led student-centered coding workshops and project-based learning sessions using open-source tools (e.g., Google Colab, GitHub) to bridge theory and practical implementation.
- Supervised semester-long student mini-projects addressing real-world datasets and social impact themes, guiding students through project scoping, model prototyping, and presentations.

- Provided mentoring and academic advising to over 10 students, enhancing student performance through targeted support in AI concepts, Python fluency, and debugging strategies.

PROFESSIONAL EXPERIENCE

Academic Resources Institute, RMUTT

AI & Data Analyst

Thailand

Jun 2023 – Sep 2023

- Engineered interactive data dashboards using Python and Tableau to surface latent trends from multi-source institutional data, informing university-level planning.
- Conducted statistical analysis and clustering to uncover performance anomalies across departments, aiding resource reallocation and efficiency monitoring.
- Automated monthly reporting pipelines and visual workflows to streamline executive communication and policy decisions.

Optice Tech LLC

Web Developer

Wyoming, USA

Feb 2020 – Jul 2023

- Built and deployed responsive, data-driven web applications with custom WordPress integrations tailored to client needs in education and e-commerce sectors.
- Integrated analytics APIs (e.g., Google Analytics, Hotjar) and applied SEO optimizations to enhance content discoverability and boost organic reach by 60%.
- Spearheaded UX enhancements through A/B testing, user feedback loops, and accessibility improvements, improving user retention and reducing bounce rates.

Heritage Solutions & Technology

IT Support Specialist

Lagos, Nigeria

2011 – 2020

- Led technical support operations for enterprise systems, resolving 95% of issues within SLA timelines through structured troubleshooting and escalation workflows.
- Delivered hands-on user training for productivity suites, cybersecurity practices, and internal platforms, increasing digital literacy among staff.
- Managed IT infrastructure upgrades and implemented preventative maintenance protocols, reducing system downtime by 40%.

PROJECTS

Causal Modeling for COVID-19 Surge Prediction Using Time-Series ML

[GitHub Repository](#)

Jan 2023 – Mar 2023

- Designed a time-series machine learning pipeline for early-stage outbreak modeling, leveraging lag-based features and exogenous indicators (mobility, testing rates) to predict COVID-19 case surges across regions.
- Compared the forecasting performance of ARIMA, Random Forest, and LSTM models using metrics such as RMSE and MAE under multiple validation windows.
- Visualized spatiotemporal prediction outputs through interactive dashboards to support epidemiological monitoring and public health planning.

Forecasting Volatility in Gold Markets via Statistical and ML Techniques

[GitHub Repository](#)

Oct 2022 – Dec 2022

- Investigated gold price volatility using autoregressive models and supervised ML regressors trained on macroeconomic and market sentiment features.

- Evaluated model robustness under rolling forecast origin evaluation; performed feature attribution to identify key economic signals driving price changes.
- Presented analytical findings through interactive visualizations, enabling interpretation of forecast patterns for financial decision-making scenarios.

SELECTED PUBLICATIONS

- **S. Nwaiwu**, et al., “The brittleness of transformer feature fusion: A comparative study of model robustness in engineering misinformation detection,” *Results in Engineering*, Elsevier, vol. 20, 2025. DOI: [10.1016/j.rineng.2025.107783](https://doi.org/10.1016/j.rineng.2025.107783)
Analyzes robustness degradation in multimodal transformer architectures under adversarial noise for misinformation classification.
- **S. Nwaiwu**, et al., “Decoding disinformation: A feature-driven explainable AI approach to multi-domain fake-news detection,” *Applied Sciences*, MDPI, vol. 15, no. 17, 2025. DOI: [10.3390/app15179498](https://doi.org/10.3390/app15179498)
Proposes a modular explainable framework for cross-domain fake-news detection using interpretable feature engineering.
- **S. Nwaiwu**, et al., “Parameter-efficient fine-tuning for low-resource text classification: A comparative study of LoRA, IA3, and ReFT,” *Frontiers in Big Data*, vol. 8, 2025. DOI: [10.3389/fdata.2025.1677331](https://doi.org/10.3389/fdata.2025.1677331)
Benchmarks parameter-efficient tuning strategies on low-resource classification tasks in NLP using BERT-based models.
- **S. Nwaiwu**, et al., “Fine-tuned BERT model for hate-speech detection in political discourse,” in *Proc. of the Int’l Conf. on ICT and Knowledge Engineering (ICT&KE)*, IEEE, 2024. DOI: [10.1109/ICTKE62841.2024.10787162](https://doi.org/10.1109/ICTKE62841.2024.10787162)
Explores bias-aware fine-tuning of BERT models on political speech datasets for hate-speech classification.

SELECTED COURSES

Master’s Courses

- Machine Learning for Data and Information Science
- Data Analytics with Statistical Modeling
- Data Visualization and Communication
- Fundamentals of Data and Information Science
- Mathematical Optimization Techniques
- Research Methodology
- Database Systems
- Digital Business Management

Bachelor’s Courses

- Artificial Intelligence
- Machine Learning
- Data Mining
- Advanced Data Analytics
- Statistical Modeling and Data Analysis
- Simulation and Modelling
- Data Visualization
- Database Management Systems

AWARDS

ECUBE Excellence Award Rajamangala University of Technology Thanyaburi, Thailand 2021
Conferred in recognition of exceptional academic performance and research excellence among top-performing students across the Faculty of Science and Technology.

TOEIC Excellence Award Rajamangala University of Technology Thanyaburi, Thailand 2022
Awarded for achieving the highest TOEIC score at both university and provincial levels, placing in the top performance percentile nationally.

ENGLISH TESTS

TOEFL iBT: 80 (overall score)

Reading: 19 | Listening: 21 | Speaking: 22 | Writing: 18

SKILLS

- **Programming:** Python (advanced; data science, ML prototyping), SQL (intermediate; query optimization, data wrangling)
- **Machine Learning:** Supervised & Unsupervised Learning, Deep Learning (CNNs, RNNs), Robust & Causal Modeling, Explainable AI (SHAP, LIME)
- **Natural Language & Multimodal Processing:** Text Classification, Transformer Architectures (BERT, GPT), Vision–Language Modeling (CLIP, ViLT)
- **Frameworks & Tools:** PyTorch, Scikit-learn, Hugging Face Transformers, Git, Jupyter, Pandas, Matplotlib, Weights & Biases
- **Research & Analytical Skills:** Experimental Design, Scientific Writing, Hypothesis Testing, Data Visualization, Reproducibility Practices

REFERENCES

Prof. Dr. Nipat Jongsawat

Professor, Faculty of Science and Technology
Rajamangala University of Technology Thanyaburi, Thailand
Academic supervisor and research mentor
Scholar Profiles: [Research Profile](#) | [Google Scholar](#)

Prof. Dr. Anucha Tungkasthan

Professor, Faculty of Science and Technology
Rajamangala University of Technology Thanyaburi, Thailand
Thesis co-advisor and academic recommender
Scholar Profiles: [Research Profile](#) | [Google Scholar](#)